



# Python for Fixed Income

COURSE 4 · MACHINE LEARNING

## Decision Trees

WEEK 2 · DAY 4

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This course is educational content created in a personal capacity. Nothing here is investment advice or a recommendation to buy or sell any security. All portfolio data is synthetic.

# A model that asks questions

## Ask

a question of one column

## Split

the accounts in two

## Stop

the hard part: today's lesson

## Allowed names

model probability of default in this file

flagged at threshold  $t$

the model's score

## Never called

a credit decision or an approval

a lending policy or a lending rule

a risk grade for a real person

**Real UCI data: 30,000 accounts, Taiwan, 2005. No person column is a feature. Test rows stay closed.**

# A tree as questions

| Word  | Meaning                             |
|-------|-------------------------------------|
| Node  | A group of training accounts        |
| Root  | The first node: every account       |
| Split | Is this value at most t? Yes left   |
| Leaf  | A node with no question             |
| Depth | Questions from root to deepest leaf |

```
|--- PAY_0 <= 1.50  
|   |--- weights: [13455.00, 2660.00] class: 0  
|--- PAY_0 > 1.50  
|   |--- weights: [563.00, 1322.00] class: 1
```

18,000 training accounts. Left: codes -2, -1, 0, 1 (two undocumented). Right: a delay of two months or more.

**PAY\_0 <= 1.5: how this file splits, in its training rows. Not a rule for lending.**

# Gini impurity, and a split by hand

$Gini = 1 - p^2 - (1 - p)^2$       $p$  = share of defaults; 0 pure, 0.5 half and half

| Group              | Accounts | Defaults | Share  | Gini   |
|--------------------|----------|----------|--------|--------|
| all training rows  | 18,000   | 3,982    | 0.2212 | 0.3446 |
| left: PAY_0 ≤ 1.5  | 16,115   | 2,660    | 0.1651 | 0.2756 |
| right: PAY_0 > 1.5 | 1,885    | 1,322    | 0.7013 | 0.4189 |

**0.2906**

weighted Gini of the two sides

**0.0539**

decrease: the largest of any question

Four counts by COUNTIFS. Excel does it too.

# The default tree overfits

**38**

questions deep

**2,750**

leaves

**6.5**

accounts per leaf

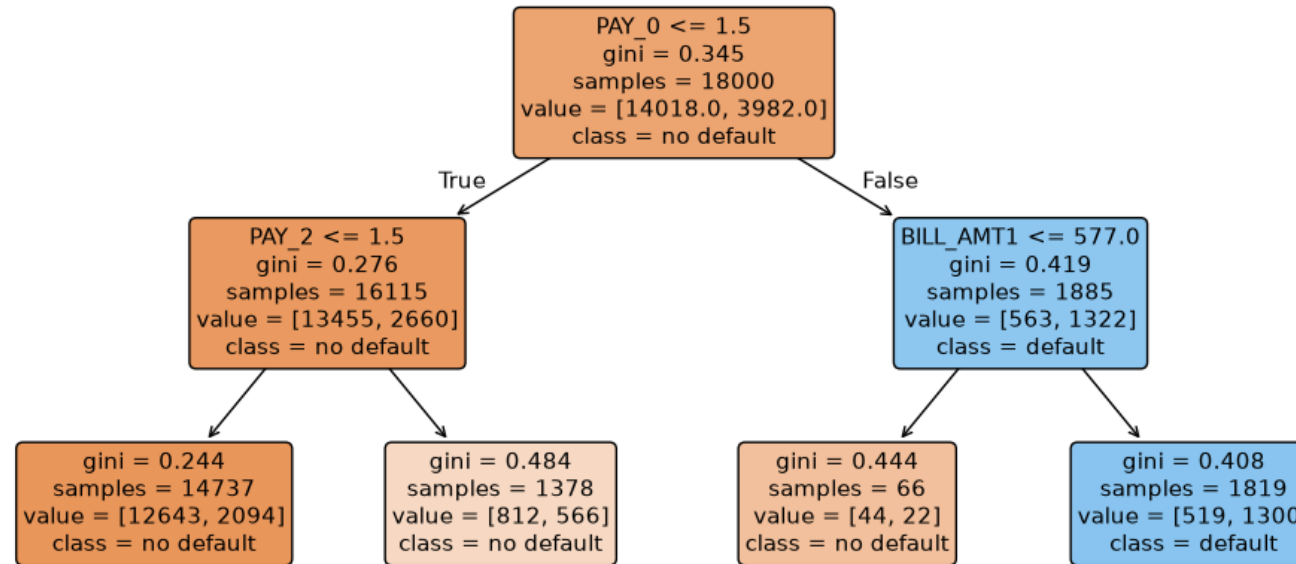
|                 | ROC AUC | Recall at 0.5 | Precision |
|-----------------|---------|---------------|-----------|
| Training rows   | 0.9997  | 0.9709        | 0.9895    |
| Validation rows | 0.6161  | 0.4032        | 0.3888    |

Training recall is not 1: 48 groups of identical accounts (413 in all) hold both outcomes.

**Nearly memorised on training rows, much worse on rows it never saw: overfitting.**

# Reading a tree: depth 2

A tree of depth 2 on the 18,000 training accounts (UCI card file)



**In this file's training rows, of accounts with status 2 or more and a bill above 577, 71.5 percent defaulted. A description, not a rule.**

# Class weights: make the rare class count

2.2602

weight of a default

0.6420

weight of a non-default

9,000

total weight of each class

| Depth 3, validation rows | ROC AUC | Recall | Precision |
|--------------------------|---------|--------|-----------|
| Unweighted               | 0.7197  | 0.3662 | 0.6676    |
| class_weight="balanced"  | 0.7491  | 0.5946 | 0.4524    |

Weighted output is a score: 0.2770 in a leaf of 8,346 accounts where 0.0981 defaulted.

# Pre-pruning: stop the tree early

| Setting          | Stops                    |
|------------------|--------------------------|
| max_depth        | a path longer than this  |
| min_samples_leaf | a leaf smaller than this |

```
cv=mlkit.cv_folds() # StratifiedKFold(5,  
                    #   shuffle=True, 42)
```

Chosen by five-fold cross-validation inside the 18,000 training rows. 38 tests pass.

**depth 8**  
leaves of at least 200

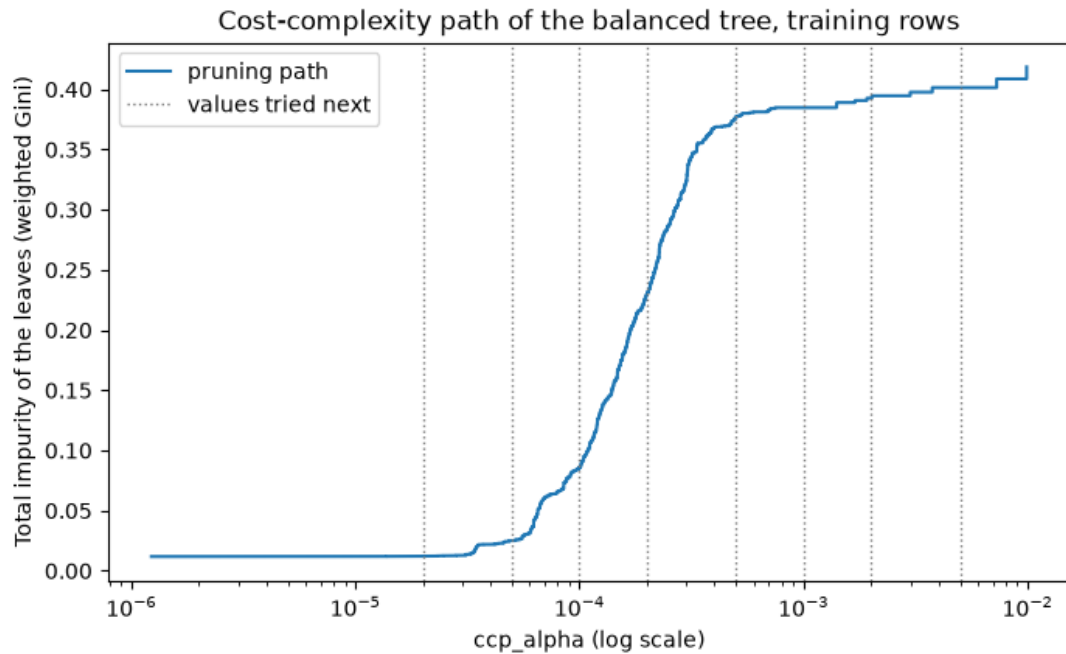
**0.7664**  
cross-validated ROC AUC

**52**  
leaves

**0.7633**  
validation ROC AUC

The validation rows stay out of the choice, so they can report it.

# Post-pruning: grow, then cut back



1,743 alphas; best ccp\_alpha 0.001 (cross-validated 0.7598).

| Validation rows       | Depth | Leaves | ROC AUC |
|-----------------------|-------|--------|---------|
| default               | 38    | 2,750  | 0.6161  |
| default, balanced     | 49    | 3,233  | 0.6158  |
| pre-pruned, balanced  | 8     | 52     | 0.7633  |
| post-pruned, balanced | 6     | 12     | 0.7627  |

Balanced trees ask  $\text{PAY}_0 \leq 0.5$  first: weights change the questions.

# Feature importance, and what it does not mean

| Column    | Impurity | Permutation |
|-----------|----------|-------------|
| PAY_0     | 0.7115   | 0.1130      |
| PAY_AMT2  | 0.0705   | 0.0108      |
| LIMIT_BAL | 0.0367   | 0.0185      |
| BILL_AMT1 | 0.0303   | 0.0030      |
| PAY_4     | 0.0293   | 0.0145      |

## Not cause

The tree used it to sort accounts

## Not direction

Higher value: more or fewer?

## Shared

Related PAY\_ columns mask each other

## Training rows

Impurity favours many split points

**Impurity: training rows. Permutation: fall in validation ROC AUC when a column is shuffled.**

# Scaling does not change a tree

A split asks above or below; scaling keeps the order of values and thresholds.

**6,000**

validation accounts, same output (pre-pruned tree)

**5**

accounts exactly on a threshold (default tree)

**Same tree, node for node. A tree needs no scaling.**

# Excel side by side: a tree by hand

## Excel side by side (tested in Excel)

`=COUNTIFS(A2:A18001,"<=1.5")`

`=COUNTIFS(A2:A18001,"<=1.5",D2:D18001,1)`

`=1-(G2/G1)^2-((G1-G2)/G1)^2`

`=G3/G1*G8+G5/G1*G9 =G7-G10`

`=AVERAGEIFS(D2:D18001,A2:A18001,">1.5",C2:C18001,">577")`

`=IF(A2<=1.5,IF(B2<=1.5,0,0),IF(C2<=577,0,1))`

`=H1/(H1+H2) =H1/(H1+H3)`

16,115 accounts on the left

2,660 defaults on the left

Gini 0.3446

0.2906, decrease 0.0539

leaf share 0.7147

the tree's flag, all rows

recall 0.3052, precision 0.6935

**Excel has no decision tree. It can count, weigh, and apply a tree Python fitted.**

# Save the day: tag the pruned trees

## 1. Code

mlkit.py and test\_mlkit.py

## 2. Unpruned

results.csv: two trees

## 3. Pruned

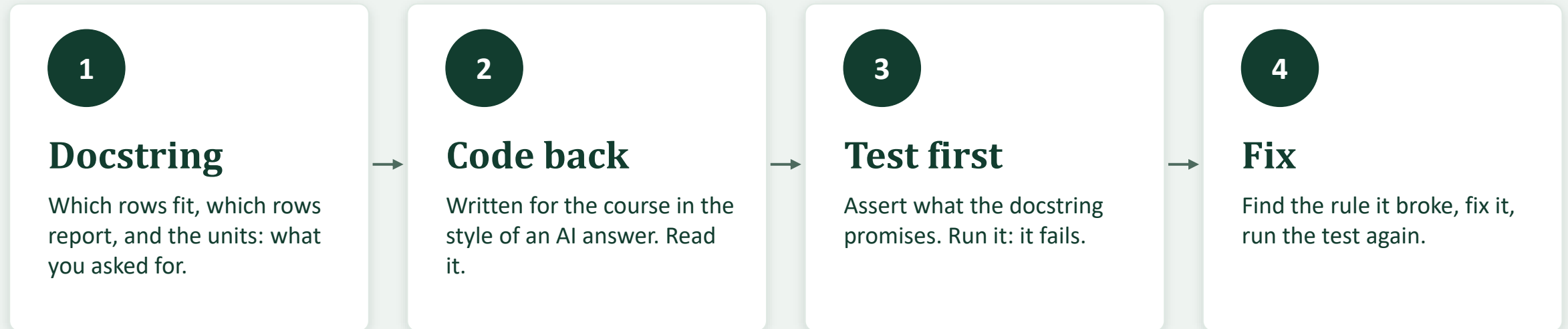
two more rows, then git tag tree-pruned

```
model,rows,roc_auc,average_precision,threshold,accuracy,precision,recall,f1
default,validation,0.6161,0.2917,0.5000,0.7278,0.3888,0.4032,0.3959
"default, balanced",validation,0.6158,0.2926,0.5000,0.7262,0.3887,0.4160,0.4019
"pre-pruned, balanced",validation,0.7633,0.5248,0.5000,0.7603,0.4682,0.6149,0.5316
"post-pruned, balanced",validation,0.7627,0.4954,0.5000,0.7683,0.4805,0.5840,0.5272
```

**A tag names one commit for good: git show tree-pruned:results.csv.**

# The AI check

The docstring is the prompt. The code that comes back runs, and has one modelling bug.



Today: the search is fitted on training and validation rows (24,000), then scored on validation: 0.7645.

*Code that runs is not the same as code that is right.*

# What you can do now

- A tree picks the question with the largest Gini decrease
- Left to grow it memorises: 0.9997 training, 0.6161 validation
- Class weights make defaults count; the output becomes a score
- Prune before or after, chosen by cross-validation on training rows
- Importance says what the tree used, never why

**Week 2, Day 5** Case study: the card file, a threshold for recall 0.70, test rows opened once